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Abstract

This paper studies how heterogeneity in trading speed and information timing affects stability in an agent-based limit-order-book model with heterogeneous traders and a market maker. The experiments vary execution speed, information timing, and order-book depth, while post-shock stability is measured after a negative market shock introduced during the simulation. Its contribution is a controlled mechanism-separation design: execution priority, market-clock representation, and information delay under thin liquidity are tested within the same limit-order-book environment. Across the speed grid, the clearest threshold appears in the speed-multiplier-2 regime: the share of fast trend-following agents produces a volatility tipping point at $\phi^* = 0.2$, with the maximum mean volatility ratio reached at $\phi = 0.4$. Delay-only settings do not produce a clean tipping point, and combined speed-delay effects are nonlinear. The calendar-time versus event-time comparison shows that the speed result is not only a market-clock artefact: calendar time and comparable event time both give $\phi^* = 0.2$, although event time is not consistently stabilising. H2 is therefore interpreted as partial support for a clock-measurement effect, not as a general stabilising effect of event time. The original volatility-clustering hypothesis is not supported after multiple-testing correction, whereas the revised shock-absorption formulation is partially supported: delay under limited order-book depth is associated with larger maximum market-price drawdown.

Аннотация

Данная работа исследует влияние неоднородности скорости исполнения и задержки информации на устойчивость рынка в агентной модели лимитного стакана с гетерогенными трейдерами и маркет-мейкером. Эксперименты варьируют скорость исполнения, режим времени и глубину лимитного стакана в разных сценариях симуляции, а устойчивость рынка оценивается после отрицательного шока, введённого в ходе симуляции. Вклад работы состоит в контролируемом дизайне разделения механизмов: приоритет исполнения, представление рыночного времени и задержка информации при тонкой ликвидности проверяются в одной и той же среде лимитного стакана. В сетке скоростных режимов наиболее чёткий порог возникает в режиме множителя скорости 2: доля быстрых тренд-следящих агентов приводит к критическому переходу волатильности при $\phi^* = 0.2$, а максимальное среднее отношение волатильности достигается при $\phi = 0.4$. Сценарии только с задержкой не дают чистой критической точки, а совместные эффекты скорости и задержки нелинейны. Сравнение календарного и событийного времени показывает, что результат скоростного эксперимента не является лишь артефактом выбранных часов: календарное время и сопоставимое событийное время дают $\phi^* = 0.2$, хотя событийное время не является универсальным стабилизатором. Поэтому H2 трактуется как частичная поддержка эффекта выбора рыночных часов, а не как общий стабилизирующий эффект событийного времени. Исходная гипотеза о кластеризации волатильности не подтверждается после поправки на множественные проверки. Пересмотренная формулировка о поглощении шока получает частичную поддержку: задержка при малой глубине лимитного стакана связана с большей максимальной просадкой рыночной цены.

Keywords

agent-based modelling; limit order book; high-frequency trading; execution speed; event time; information delay; market instability; shock absorption.

1 Introduction

Electronic markets can become unstable when some traders observe information and act on it faster than others. This paper studies that problem in an agent-based limit-order-book model. It focuses on three mechanisms that can affect post-shock stability: execution speed, market-clock choice, and information delay under limited order-book depth.

The empirical motivation is the class of market disruptions associated with automated trading and rapid liquidity withdrawal, including the 2010 Flash Crash [1, 2]. The paper does not try to reconstruct that event. Instead, it uses a controlled simulation to separate mechanisms that are difficult to isolate in real market data: which agents act first, how market activity is measured over time, and whether delayed reaction becomes more damaging when the book is thin.

The first experiment, referred to as H1 below, tests execution speed. The parameter ϕ denotes the share of chartists assigned to the fast trend-following group. Fast agents are called before slow agents, and the speed multiplier controls how many trading calls fast agents receive before slow agents trade. The speed grid tests multiplier values $\{1, 2, 3, 5\}$ while varying ϕ . The final clean speed result is reported for the speed-multiplier-2 regime with no information lag. A volatility tipping point is defined as the first value of ϕ at which the mean post-shock volatility ratio exceeds $1.3\times$ the $\phi = 0$ baseline.

The second experiment, referred to as H2 below, compares calendar time with executed-volume event time. Calendar time uses equal clock intervals, but those intervals can contain different amounts of executed trading activity. Event time instead groups market observations by executed volume. The aim is to test whether the speed-driven tipping boundary is preserved or shifted when market activity is measured by executed volume. The comparison therefore treats the market clock as part of the measurement problem, without assuming that event time is always more stable. In the final interpretation, comparable event time preserves the H1 tipping boundary, but other event-time thresholds shift that boundary and the paired equal-volume tests show only limited corrected reductions in measured instability. H2 therefore receives partial support as a clock-measurement effect, not as evidence that event time is generally stabilising.

The third experiment, referred to as H3 below, concerns information delay under limited order-book depth. The original hypothesis expected delay under such conditions to generate stronger volatility clustering. The statistical tests do not support that broad volatility-clustering claim. The final interpretation is therefore narrower: delay under thin-order-book conditions is evaluated as weaker shock absorption, measured mainly by maximum market-price drawdown. Maximum market-price drawdown means the largest relative fall from the pre-shock market price to the lowest post-shock market price.

The paper’s contribution is to use the ABM as a controlled mechanism-separation laboratory.

Rather than proposing a new market model, it holds one limit-order-book environment fixed and varies three channels in turn: execution priority and speed heterogeneity, calendar-time versus event-time representation, and information delay under thin liquidity. This design makes the mixed findings interpretable rather than anomalous: speed-driven tipping appears in a clean regime, event time changes measurement without uniformly stabilising the market, and delay-depth effects are strongest for maximum market-price drawdown rather than broad volatility clustering. Methodologically, the design combines a fixed market environment with Monte Carlo validation, non-parametric and corrected inference, and robustness checks; substantively, it identifies which timing and liquidity claims survive validation and which claims were too broad.

This study makes three specific contributions. First, it separates execution speed, market-clock representation, and information delay under thin liquidity in one fixed limit-order-book ABM. Second, it validates H1–H3 using Monte Carlo simulation, non-parametric tests, bootstrap uncertainty, and multiple-testing correction. Third, it distinguishes robust mechanisms from claims that are too broad: H2 receives partial support only as a clock-measurement effect, while H3 receives partial shock-absorption support only through maximum drawdown.

2 Literature Review

The first group of studies concerns liquidity crises and the behaviour of fast traders during market stress. Official and academic analyses of the 2010 Flash Crash generally do not treat high-frequency traders as the sole cause of the event. They instead describe the crash as the outcome of interacting factors, including a large sell order executed by an automated trading algorithm, high-frequency trading activity, market-making behaviour, and rapidly changing liquidity provision [1, 2]. This interpretation motivates studying speed and liquidity together rather than treating price dynamics as the outcome of a single trader type. Easley, López de Prado, and O’Hara’s Flash Crash paper is used here for the liquidity-crash interpretation and flow-toxicity framing: one-sided or adverse order flow can make liquidity provision riskier and may encourage liquidity suppliers to reduce posted depth or widen spreads [3]. Their later flow-toxicity and volume-clock papers frame VPIN, high-frequency toxicity, and volume-time measurement, but these tools are treated as a conceptual measurement language rather than as settled crash predictors [4, 5]. Andersen and Bondarenko’s critiques are therefore important: they show that VPIN-style toxicity measures should not be treated as settled early-warning predictors [6, 7]. For this reason, flow toxicity is used here only as a conceptual language for adverse order flow and liquidity withdrawal, not as a statistic that the model tries to validate.

A second strand treats speed advantage as a strategic and market-design variable. The high-

frequency trading arms-race literature suggests that very small timing advantages can change priority, incentives, and the distribution of gains from reacting first [8, 9]. Studies of fragmented markets show that latency advantages can reward faster reaction [10], while exchange-latency models show that the welfare and liquidity effects of speed are regime-dependent rather than uniformly destabilising [11]. Evidence that machine-speed trading can operate below human reaction time also frames the modelling choice to distinguish fast and slow agents [12]. These studies motivate execution priority as an economically meaningful mechanism, while leaving its net stability effect as an empirical or simulation question.

The event-time literature motivates the market-clock comparison. Volume-clock arguments emphasise that market information is not distributed uniformly across calendar time: equal intervals of calendar time may correspond to very different amounts of executed trading activity [5]. This suggests that a volatility pattern measured in calendar ticks may partly reflect the chosen clock rather than only the underlying trading mechanism. For this reason, H2 compares calendar time with executed-volume event time as a market-clock question. The relevant issue is whether event time preserves or shifts the speed-driven tipping boundary, not whether it is universally better or uniformly stabilising.

Agent-based and heterogeneous-agent models provide the modelling logic for the paper. Such models are useful when aggregate instability, volatility persistence, or liquidity stress may arise from heterogeneous rules, local feedback, inventory constraints, and market-microstructure details rather than from a representative trader. Bookstaber and Paddrik provide an agent-based modelling framework for crisis-liquidity dynamics [13]. Kirman and Teyssi re show that interacting-agent models can reproduce persistent volatility patterns in financial time series [14]. Recent survey work similarly frames agent-based modelling as a way to connect micro-level behaviour with macro-level financial outcomes while retaining explicit heterogeneity [15]. This literature does not determine the direction of the paper’s results. It motivates the experimental design: the market model is held fixed, heterogeneous trader types and a market maker are retained, and one timing or liquidity mechanism is changed at a time.

A related literature studies liquidity depth, strategic trading under stress, delayed reaction, and volatility clustering. Predatory-trading and episodic-liquidity-crisis models are relevant because they show how limited absorption capacity can amplify stress when traders demand liquidity strategically or when liquidity is thin [16, 17]. These papers are not direct models of information delay; rather, they frame why thin order books can make shocks harder to absorb. Delay-based agent models add a closer timing mechanism. Zhou et al. analyse information delay and liquidity in a delay-based agent model and show that the effect of delay on stability can be nonlinear rather than simply destabilising [18]. This prior work motivates testing delay and liquidity jointly, but it also supports the cautious design used here: the model does not assume in advance that delay must generate instability, and the final H3 interpretation distinguishes the rejected volatility-clustering claim from the narrower shock-absorption result.

Regulatory work on technological change in markets also frames the project. Market-integrity concerns raised by automation and high-speed trading are not limited to individual strategies; they include how market design, information timing, order priority, and liquidity provision interact under stress [19]. This regulatory perspective is useful because it treats instability as a property of market design rather than only of individual trader behaviour. This perspective is consistent with treating the three experiments as related but separate tests. The speed experiment changes execution priority, the clock comparison changes the measurement of market activity, and the delay-depth experiment changes the conditions under which shocks are absorbed. None of these design choices is assumed to have a uniform stabilising or destabilising effect across regimes.

Overall, the literature motivates the mechanisms studied in the simulations but does not pre-determine the findings. Prior work suggests plausible channels through which speed, event time, delay, and order-book depth can matter, while also showing that their effects depend on market design and liquidity conditions. Mixed findings are therefore informative rather than anomalous: they help identify which claims are robust in this particular model, which are sensitive to the market-clock specification, and which are too broad relative to the available evidence.

Table 2.1 summarises the literature as a mechanism map rather than as a simple citation list: each source is linked to the specific channel tested in the simulations.

Table 2.1 – Representative literature and relevance to this study

Source	Mechanism or topic	Main contribution	Relevance to this project
CFTC/SEC Staff Report; Kirilenko et al. [1, 2]	Flash Crash and liquidity crisis	Market disruption emerged from interacting selling pressure, automated trading, liquidity provision, and market-maker behaviour.	Motivates studying speed and liquidity together rather than blaming one trader type.
Easley, López de Prado, and O’Hara [3, 4]	Flow toxicity	Adverse or one-sided order flow can make liquidity provision risky.	Motivates interpreting post-shock order flow and liquidity withdrawal.
Andersen and Bondarenko [6, 7]	VPIN critique and toxicity-measure debate	Early-warning toxicity measures are contested.	Justifies using toxicity conceptually rather than treating one statistic as a settled crash predictor.
Budish, Cramton, and Shim; Biais, Foucault, and Moinas [8, 9]	Speed advantage and strategic fast trading	Small timing advantages can affect priority, rents, and market design.	Motivates H1 execution-speed heterogeneity.
Wah and Wellman [10]	Latency arbitrage in fragmented markets	Agent-based analysis represents strategic latency advantage in a fragmented market setting.	Supports modelling speed as an agent-level mechanism.
Menkveld and Zoican [11]	Exchange latency and liquidity	Lower latency can improve or worsen liquidity depending on market conditions.	Supports a regime-dependent interpretation of speed effects.
Johnson et al. [12]	Machine-speed market ecology	Market events can occur below human reaction time.	Motivates separating fast and slow agents.
Easley, López de Prado, and O’Hara [5]	Volume and event time	Volume-clock metrics frame high-frequency decisions in units of traded volume rather than equal calendar-time intervals.	Motivates the H2 calendar-time versus event-time comparison.
Bookstaber and Paddrik [13]	ABM crisis-liquidity dynamics	ABM can represent liquidity stress and crisis dynamics through agent interaction.	Supports using a limit-order-book ABM.
Kirman and Teyssi�re [14]	Interacting agents and volatility persistence	Heterogeneous interaction can generate persistent volatility patterns.	Motivates the original H3 volatility-clustering formulation.
Axtell and Farmer [15]	ABM in economics and finance	ABM connects micro-level rules with macro-level outcomes under heterogeneity.	Supports the general modelling approach.
Brunnermeier and Pedersen; Carlin, Lobo, and Viswanathan [16, 17]	Liquidity stress and predatory trading	Limited absorption capacity and strategic liquidity demand can amplify stress.	Motivates the thin-book shock-absorption logic.
Zhou, Zhong, and Li [18]	Information delay and liquidity in a delay ABM	Delay and liquidity can affect market stability in an agent-based setting.	Provides a direct prior motivation for testing information delay and liquidity jointly in an ABM, though not specifically for thin-book shock absorption.
IOSCO [19]	Technological change and market integrity	Automation and high-speed trading create market-integrity and design concerns.	Supports the policy relevance of separating speed, clock choice, and liquidity mechanisms.

Taken together, these studies motivate the three channels used here, but they do not test execution speed, market-clock representation, and information delay under thin liquidity as separate mechanisms inside one fixed LOB ABM with corrected statistical validation.

The model section translates this mechanism map into explicit experimental variation: speed is tested through execution priority, event time through the market clock, and delay-depth interaction through post-shock liquidity absorption.

Accordingly, the paper does not estimate a real-market causal effect; it uses a controlled ABM to separate mechanisms that are usually observed together in real market data.

3 Model and Methodology

3.1 Limit-Order-Book Architecture

The model simulates a stylised limit-order-book market for a representative financial instrument. The book has two sides: a bid side, where buy limit orders are ranked by price priority with the best bid first, and an ask side, where sell limit orders are ranked by price priority with the best ask first. Trades occur when compatible buy and sell orders cross, so transaction formation is governed by the interaction between submitted orders and the current state of the book.

When both sides of the book are available, the market price is represented by the midpoint between the best bid and the best ask. The spread measures the gap between these two quotes, while book depth captures the amount of available liquidity behind quoted prices. These variables are recorded together with returns, order-book state, and market-maker stress indicators, and they are later used to interpret post-shock stability and liquidity conditions.

The order-book environment should be interpreted as a controlled modelling laboratory rather than as a calibrated replica of a particular exchange. This design makes it possible to vary execution speed, information timing, and liquidity conditions while keeping the rest of the market mechanism fixed.

3.2 Agent Types and Market Maker

The simulated market contains heterogeneous trader types. Random traders provide background order flow and noise, preventing the book from being driven only by directional strategies. Fundamentalists trade against deviations from an estimated fundamental value and therefore provide a stabilising pressure when prices move away from that benchmark. Baseline chartists follow sentiment and price-related signals, which gives the model a behavioural component distinct from fundamental-value trading.

For H1, the experiments use a trend-following chartist variant because the original chartist rule was not designed to represent crash-amplifying fast trend followers. Fast trend-following chartists can react earlier after a negative shock and sell into the falling market, making them the relevant group for studying execution-speed heterogeneity. Universalists are supported by the framework and can switch between fundamentalist and chartist behaviour, but they are not central to the final experiments.

The market maker supplies two-sided liquidity subject to inventory constraints. This role is important because post-shock stability depends not only on directional trader behaviour, but also on whether normal liquidity provision can be maintained. Market-maker stress or panic is therefore relevant for the stability measurement: failure to provide normal two-sided liquidity is treated as part of the market's post-shock response.

3.3 Shock Design and Common Simulation Setup

Each simulation follows the same basic sequence: agents update behavioural states, submit orders, the exchange matches compatible orders, and the information object records market state. All experiments use a controlled negative market shock inside the simulation rather than an externally calibrated market episode. The shock occurs at tick 200 and shifts standing order-book quotes downward by 10 price units, after which the post-shock period is used to measure the market’s stability response.

The baseline simulation horizon is 500 iterations. Repeated Monte Carlo runs are used throughout, with run counts varying by experiment as reported in Table 3.1. The common setup keeps the interpretation of each hypothesis tied to a local mechanism rather than to a different market model.

The common setup also clarifies how the three hypotheses are separated. For H2, calendar-time and event-time runs preserve the same behavioural environment when testing the effect of the market clock. For H3, book depth and information delay are varied to test shock absorption under different liquidity conditions.

Table 3.1 – Common experimental setup.

Component	Specification
Simulation horizon	500 iterations
Shock	Negative market shock at $t = 200$ with $\Delta p = -10$
Population	10 chartists, 10 fundamentalists, 5 random traders, and 1 market maker
Main H1 grid	$\phi \in \{0.0, 0.1, \dots, 1.0\}$; speed multiplier $\in \{1, 2, 3, 5\}$ in the speed-only branch
Monte Carlo runs	H1 and H2 calibrated tipping: 30 runs per cell; H2 equal-volume: 50 paired runs per ϕ ; H3 confirmatory: 200 runs per cell
Main metrics	Volatility ratio, spread ratio, maximum market-price drawdown, recovery time, and market-maker stress
Tipping rule	First ϕ where mean volatility ratio crosses $1.3\times$ the $\phi = 0$ baseline

Table 3.2 – Mechanism-to-experiment map.

Mechanism	Parameter varied	Main evidence	Final interpretation
Execution speed	ϕ , speed_multiplier	Volatility ratio and $1.3\times$ tipping rule	H1 is supported only in the speed-multiplier-2 regime; the effect is not monotonic.
Market clock	Calendar time; event-time thresholds V6, V13, V19	Tipping boundary and equal-volume paired tests	H2 gives partial clock-measurement robustness support and weakens the calendar-time artefact concern; not general stabilisation.
Delay $\times$ thin book	info_lag, book_volume	Volatility-clustering tests and shock-absorption metrics	Original H3 is not supported; revised H3 is partially supported through maximum market-price drawdown.

Hypothesis operationalisation. H1 tests execution-speed heterogeneity by varying the fast-chartist share ϕ and the speed multiplier, using the post-shock volatility ratio and the $1.3\times$ tipping rule as the primary decision rule. H2 tests market-clock measurement by varying calendar time versus executed-volume event time, using tipping-boundary preservation and paired equal-volume differences as evidence; H2 is interpreted as partial clock-measurement robustness support when the calibrated event-time clock preserves the H1 boundary and weakens the calendar-time artefact concern, while the absence of uniform stabilisation rules out a broad event-time stabilisation claim. H3 tests the delay $\times$ thin-book mechanism by varying `info_lag` and `book_volume`; the original formulation would require corrected support in volatility-clustering metrics, while the revised formulation would require corrected deterioration in shock-absorption metrics.

3.4 Metrics

The main stability metrics compare behaviour after the shock with benchmarks measured before the shock. The volatility ratio is computed as a post-shock price volatility measure divided by the pre-shock price volatility benchmark. The spread ratio is computed analogously as the post-shock relative spread divided by the pre-shock relative spread. These ratios make the interpretation focus on the change in market conditions caused by the shock rather than on the absolute level of prices or spreads.

Maximum market-price drawdown is the largest relative fall from the pre-shock market price to the lowest post-shock market price. Recovery time is the number of ticks until the price returns close to the pre-shock level; if no recovery occurs, it is set to the remaining post-shock horizon. In the implementation, “close to the pre-shock level” means that the market price enters a 2 percent band around the pre-shock price. This is an operational tolerance, not a theoretical market constant. It avoids treating tiny tick-level fluctuations in the simulated limit order book as economically meaningful non-recovery, while still requiring the price to return very near its pre-shock benchmark. Market-maker stress is measured by the panic ratio, defined as the fraction of post-shock ticks in which the market maker cannot provide normal two-sided liquidity. This can occur when no valid spread is available or when inventory constraints make feasible bid or ask quote volume unavailable.

The H2 event-time comparison uses equal-volume metrics to separate measured instability from trading intensity. These metrics normalise realised variation by executed volume or trade count, so the calendar-time and event-time comparison does not only reflect the fact that different clock choices can contain different amounts of trading activity.

For H3, the original volatility-clustering formulation uses metrics such as the autocorrelation of absolute returns, volatility persistence, and the share of high-volatility clusters. The revised shock-

absorption formulation instead focuses on recovery time, maximum market-price drawdown, and the stabilisation gap. This distinction separates persistent volatility patterns from the question of how deeply and durably the market moves after the shock.

3.5 Formal definition of stability metrics

Let p_t denote the market price at simulation tick t , let T denote the number of recorded price observations, and let t_s denote the shock tick. The last pre-shock benchmark price is p_{t_s-1} . The pre-shock and post-shock tick sets are denoted by $\mathcal{T}_{pre}$ and $\mathcal{T}_{post}$. A small numerical constant $\varepsilon > 0$ is used in ratio denominators to avoid division by zero.

Main price-level volatility ratio.

The main H1 and calibrated H2 tipping metric is based on rolling volatility of the market-price level, not on return volatility. For a rolling window of length w , define

$$\sigma_j^p(w) = \text{sd}(p_j, p_{j+1}, \dots, p_{j+w-1}). \quad (1)$$

The implementation uses separate pre-shock and post-shock rolling-window start sets. Let

$$\mathcal{J}_{pre} = \{j : 0 \leq j \leq t_s - w - 1\}, \quad \mathcal{J}_{post} = \{j : t_s \leq j \leq T - w - 1\}. \quad (2)$$

The volatility ratio is

$$\text{VolRatio} = \frac{\frac{1}{|\mathcal{J}_{post}|} \sum_{j \in \mathcal{J}_{post}} \sigma_j^p(w)}{\frac{1}{|\mathcal{J}_{pre}|} \sum_{j \in \mathcal{J}_{pre}} \sigma_j^p(w) + \varepsilon}. \quad (3)$$

This is the metric used for the H1 tipping-point result and for the H2 calibrated calendar/event-time tipping comparison.

Spread ratio.

If a_t and b_t denote the best ask and best bid, the relative spread is

$$s_t = \frac{a_t - b_t}{p_t}. \quad (4)$$

The spread ratio is

$$\text{SpreadRatio} = \frac{\frac{1}{|\mathcal{T}_{post}|} \sum_{t \in \mathcal{T}_{post}} s_t}{\frac{1}{|\mathcal{T}_{pre}|} \sum_{t \in \mathcal{T}_{pre}} s_t + \varepsilon}. \quad (5)$$

Ticks with no valid two-sided spread are excluded from the relative-spread average.

Maximum market-price drawdown.

Maximum market-price drawdown is measured relative to the last pre-shock price:

$$\text{MDD} = \frac{p_{t_s-1} - \min_{t \geq t_s} p_t}{p_{t_s-1}}. \quad (6)$$

Higher values indicate a deeper post-shock price fall.

Recovery time.

Recovery time is the first post-shock tick at which the market price returns to a band around the last pre-shock price:

$$\text{RecoveryTime} = \min \left\{ k \geq 0 : \frac{|p_{t_s+k} - p_{t_s-1}|}{p_{t_s-1}} < \tau \right\}. \quad (7)$$

If the condition is never reached, recovery time is set to the remaining post-shock horizon. In the reported experiments, $\tau = 0.02$, so recovery means returning to a 2 percent band around the pre-shock price.

Market-maker panic ratio.

The market-maker panic ratio is

$$\text{MMPanicRatio} = \frac{1}{|\mathcal{T}_{post}|} \sum_{t \in \mathcal{T}_{post}} \mathbf{1}\{\text{at least one market maker is in panic at } t\}. \quad (8)$$

In the implementation, a market maker enters panic when no valid spread is available or when inventory constraints prevent normal two-sided quoting.

H2 equal-volume return-based metrics.

The H2 equal-volume tests use return-based realised variation metrics. These are separate from the main price-level VolRatio. The simple return is

$$r_t = \frac{p_{t+1} - p_t}{p_t}. \quad (9)$$

Let q_t be executed volume and n_t be the executed trade count at tick t . For a tick set A , define

$$\text{RV}(A) = \sum_{t \in A} r_t^2, \quad Q(A) = \sum_{t \in A} q_t, \quad N(A) = \sum_{t \in A} n_t. \quad (10)$$

The equal-volume design constructs a post-shock window A_{post} and a pre-shock benchmark window A_{pre} with comparable executed volume. The reported H2 equal-volume metrics are

$$\text{RVPerVolume} = \frac{\text{RV}(A_{post})}{\max(Q(A_{post}), 1)}, \quad (11)$$

$$\text{RVPerTrade} = \frac{\text{RV}(A_{\text{post}})}{\max(N(A_{\text{post}}), 1)}, \quad (12)$$

$$\text{VolPerSqrtVolume} = \frac{\text{sd}(\{r_t : t \in A_{\text{post}}\})}{\sqrt{\max(Q(A_{\text{post}}), 1)}}, \quad (13)$$

and

$$\text{EqualVolumeVolRatio} = \frac{\text{RV}(A_{\text{post}}) / \max(Q(A_{\text{post}}), 1)}{\text{RV}(A_{\text{pre}}) / \max(Q(A_{\text{pre}}), 1) + \varepsilon}. \quad (14)$$

Paired H2 differences are computed as

$$\Delta M = M_{\text{event time}} - M_{\text{calendar time}}. \quad (15)$$

For instability metrics, negative paired differences mean lower measured instability under event time.

Original H3 volatility-clustering diagnostics.

The original H3 formulation uses return-based clustering metrics. These metrics are reported for the original H3 diagnostic family, which is tested and rejected; the revised H3 interpretation uses the shock-absorption metrics below. For post-shock returns, the autocorrelation of absolute returns at lag ℓ is

$$\text{ACFAbsRet}_\ell = \text{corr}(|r_t|, |r_{t-\ell}|), \quad \ell \in \{1, 5\}. \quad (16)$$

Rolling return volatility is computed over a return window of length w . Volatility persistence is the first-order autocorrelation of post-shock rolling return volatility. The high-volatility cluster share is the fraction of post-shock rolling-volatility observations above the 90th percentile of the pre-shock rolling-volatility distribution.

H3 shock-absorption metrics.

The revised H3 shock-absorption formulation uses recovery time, maximum drawdown, and stabilisation gap. Stabilisation price is the mean market price over the final stabilisation window of the simulation:

$$\text{StabilisationPrice} = \frac{1}{L} \sum_{t=T-L}^{T-1} p_t, \quad (17)$$

with $L = 50$ in the implementation. The stabilisation gap is

$$\text{StabilisationGap} = 1 - \frac{\text{StabilisationPrice}}{p_{t_s-1}}. \quad (18)$$

Higher stabilisation-gap values mean the market settles further below the pre-shock price.

Operational tipping point.

For the H1 and calibrated H2 tipping analysis, the operational tipping point is defined using the main price-level volatility ratio:

$$\phi^* = \min \{ \phi > 0 : \overline{\text{VolRatio}}(\phi) \geq 1.3 \overline{\text{VolRatio}}(0) \}. \quad (19)$$

This is an operational threshold for comparing simulation regimes, not a structural market constant.

H3 interaction model.

For the H3 delay-by-thin-book validation, the interaction model is estimated separately by ϕ and metric:

$$\begin{aligned} y_{i,r} = & \beta_0 + \beta_1 \text{Delay}_i + \beta_2 \text{ThinBook}_i \\ & + \beta_3 (\text{Delay}_i \times \text{ThinBook}_i) + \gamma_r + \varepsilon_{i,r}, \end{aligned} \quad (20)$$

where γ_r denotes run fixed effects. The coefficient β_3 measures whether information delay and thin order-book depth jointly change the shock-absorption outcome beyond their separate effects.

3.6 Statistical Testing Strategy

The simulation outputs are Monte Carlo distributions of market-stability metrics rather than single deterministic observations. These distributions can be skewed, non-normal, heteroskedastic, and affected by outlying runs in which liquidity withdrawal or market-maker stress becomes unusually severe. For this reason, the report does not rely only on parametric tests based on normality assumptions. The statistical strategy combines operational threshold rules, non-parametric comparisons, robust regression checks, and bootstrap confidence intervals so that inference remains tied to the simulated market behaviour.

For H1, the primary decision rule is the volatility tipping rule. A tipping point is defined as the first value of ϕ where the mean volatility ratio reaches at least $1.3 \times$ the $\phi = 0$ baseline within the same regime. The $1.3 \times$ cutoff is an operational threshold for an economically meaningful post-shock volatility increase, not a theoretical constant. Mann–Whitney U tests are used as supporting validation when positive- ϕ regimes are compared against the $\phi = 0$ baseline, because the test does not require normally distributed Monte Carlo outcomes. Appendix B checks sensitivity to alternative cutoffs from $1.1 \times$ to $1.5 \times$. Bootstrap confidence intervals are used to show Monte Carlo uncertainty in the plotted metrics.

For H2, calendar-time and event-time results are compared first through the estimated tipping boundary ϕ^* . This checks whether the speed-driven threshold is preserved or shifted when observations are grouped by executed volume rather than by calendar ticks. The equal-volume post-shock tests then use

paired simulations with the same ϕ and seed while matching post-shock executed volume across paired observations. For each pair, the difference is computed as the event-time metric minus the calendar-time metric, so a negative difference means that event time gives a lower instability metric. The Wilcoxon signed-rank test is used for paired differences when the magnitude of paired changes matters, while the sign test checks only whether negative differences are more frequent than positive differences. Bootstrap confidence intervals are used for paired mean differences.

For H3, the original volatility-clustering formulation is evaluated through interaction tests over clustering metrics. The revised shock-absorption formulation is evaluated through a best-case versus worst-case comparison, where the best case represents no delay with a deeper book and the worst case represents delay with a thinner book. An OLS interaction model with HC3 robust standard errors is used to test the delay $\times$ thin-book interaction while reducing sensitivity to heteroskedasticity across simulation cells. A Mann–Whitney worst-vs-best comparison is also used to test whether the worst-case regime is systematically worse than the best-case regime without assuming normality.

Because the analysis compares several metrics, regimes, and values of ϕ , uncorrected p -values are not treated as final evidence on their own. Benjamini–Hochberg correction is used to control the false discovery rate across related tests, while Holm correction is stricter and controls the familywise error rate. Corrected results are therefore treated as stronger evidence than uncorrected results, and uncorrected findings are interpreted mainly as diagnostic or suggestive patterns.

3.7 Experimental Protocols

The experimental protocols isolate three mechanisms while keeping the common shock design and baseline market architecture fixed. In the simulations, execution speed and information delay are modelled through different channels. Execution speed is represented by activation order and by the speed multiplier: fast agents trade before slow agents and may receive additional trading opportunities within one tick. Information delay is represented by the information-lag parameter: delayed agents base their decisions on older recorded prices and spreads. This separation makes it possible to distinguish execution priority from the age of the information used for decisions.

Each protocol changes one part of the simulated environment while holding the rest of the behavioural setup as fixed as possible. H1 varies speed and delay mechanisms among trend-following chartists, H2 changes the market clock while preserving the same agent behaviour, and H3 varies delay and order-book depth to study liquidity-dependent shock absorption. The protocols therefore define the design of the experiments; the empirical outcomes are reported separately in the results section.

3.7.1 H1: Speed-Driven Tipping

For H1, the design tests whether execution-speed heterogeneity among trend-following chartists can create a post-shock instability threshold. The parameter ϕ is the share of fast chartists among chartists, not among all agents. Fast chartists act before slow chartists, and the speed multiplier gives fast chartists additional trading opportunities per tick. The trend-following chartist variant follows the agent description above; here the protocol isolates how activation priority and within-tick trading frequency affect post-shock instability.

The H1 protocol has three design branches. The speed-only branch varies both ϕ and the speed multiplier over $\{1, 2, 3, 5\}$ while keeping information lag at zero, isolating execution priority and trading-frequency advantage. Lower and moderate multipliers identify clean execution-speed tipping, while multiplier 5 is retained as an extreme diagnostic branch because very frequent fast-agent trading may move the model toward nonlinear order-book absorption or liquidity-stress behaviour. The delay-only branch varies ϕ and information lag with no speed advantage, isolating the effect of stale information. The combined branch varies both speed and delay, allowing the two mechanisms to interact within the same order-book environment.

The tipping rule is applied within each regime relative to that regime's own $\phi = 0$ baseline. A tipping point is therefore not treated as a universal boundary across all parameter settings, but as an operational marker of a large post-shock volatility increase within a specific design branch. This structure avoids interpreting speed effects as universal monotonic effects.

3.7.2 H2: Calendar Time versus Event Time

The H2 experiment was developed as a clock-representation robustness test. The initial standalone event-time prototype introduced a volume clock, but its calendar-time baseline did not reproduce the final clean H1 speed-grid result. For this reason, the final H2 specification was implemented in the same unified environment as the clean H1 benchmark. The population, behavioural rules, shock design, fast-chartist-share grid, information lag, and speed setting are held fixed at the clean H1 benchmark: `speed_multiplier=2` and `info_lag=0`. H2 then varies only the time representation, comparing calendar-time updating with event-time updating under executed-volume thresholds. This design tests whether the H1 tipping boundary is preserved or shifted when market activity is measured by traded volume rather than by fixed simulation ticks, isolating the clock effect without changing trader behaviour.

Calendar time records one observation per simulation iteration. Event time instead records observations when accumulated executed volume reaches a threshold. The V6, V13, and V19 specifications are alternative executed-volume thresholds, with V13 serving as the closest calibrated analogue to the

calendar-time baseline. The threshold comparison checks whether the same speed-driven boundary is preserved or shifted when market activity is measured by executed volume.

The equal-volume paired design provides a second H2 check. It compares calendar-time and event-time runs with the same ϕ and random seed, meaning the same deterministic random-number initialisation, while holding post-shock executed volume equal across the paired observations. This separates the clock effect from differences in total trading activity and makes the comparison focus on the measurement scale rather than on how much trading happened after the shock.

3.7.3 H3: Delay, Thin Liquidity, and Shock Absorption

H3 is organised in two analytical layers. The original broad formulation was the initial test: it varies information delay and liquidity conditions and evaluates volatility-clustering metrics. After that layer was run, multiple-testing correction showed no robust support for the broad volatility-clustering claim, so the same delay-depth mechanism was narrowed ex post to a shock-absorption framing. The revised formulation is therefore a post-hoc refinement of the same mechanism, not a separate unrelated hypothesis; it compares a best-case regime with no information delay and a deep book against a worst-case regime with one-tick delay and a thin book.

The parameter `info_lag` delays the price and spread information observed by slow agents, while `book_volume` controls order-book depth. Lower book volume represents thinner liquidity and weaker shock-absorption capacity because fewer orders are available to absorb pressure after the shock. This design tests whether delayed reaction becomes more damaging when the book is thin.

The two H3 layers use different measurement targets. The original volatility-clustering layer focuses on persistence in volatility after the shock. The revised shock-absorption layer focuses on recovery time, maximum market-price drawdown, and the stabilisation gap. This distinction separates the initial volatility-persistence claim from the narrower shock-severity and absorption refinement.

4 Empirical Results

The results are presented in three layers. First, the main subsections report the primary evidence for H1, H2, and H3. Second, the statistical-validation subsections test whether the visual and threshold-based patterns survive non-parametric, paired, or corrected inference. Third, the appendix documents robustness checks, implementation details, and output maps. This structure separates the main claims from supplementary diagnostics while keeping the full computational scope transparent.

4.1 H1: Speed-Driven Tipping

H1 concerns whether heterogeneous speed and information conditions generate a volatility tipping point after the shock. The experiment is therefore considered through three H1 branches: speed-only, delay-only, and combined speed-delay. The speed-only branch is the primary branch for the final H1 tipping claim because it isolates execution speed: information lag is fixed at zero, while execution-speed heterogeneity varies through ϕ and the speed multiplier. Within this branch, the full speed-only grid covers multipliers $\{1, 2, 3, 5\}$. Among these regimes, `speed_multiplier=2` provides the clearest and most interpretable volatility tipping pattern, so it is used as the main H1 result. Higher multipliers, especially 5, are retained as extreme diagnostic evidence of nonlinear order-book stress rather than treated as the main result. At $\phi = 0$, the baseline volatility ratio is 1.746, so the $1.3\times$ tipping threshold is 2.270. The first crossing occurs at $\phi^* = 0.2$. The maximum mean volatility ratio is 3.392 at $\phi = 0.4$. The final H1 speed-only result is reported in Table 4.1 and Figure 4.1.

Table 4.1 – Final H1 speed-only result for speed multiplier equal to 2.

Quantity	Value
Baseline volatility ratio at $\phi = 0$	1.746
Tipping threshold, $1.3\times$ baseline	2.270
Tipping point ϕ^*	0.2
Maximum mean volatility ratio	3.392 at $\phi = 0.4$
Clean speed regime	speed multiplier equal to 2

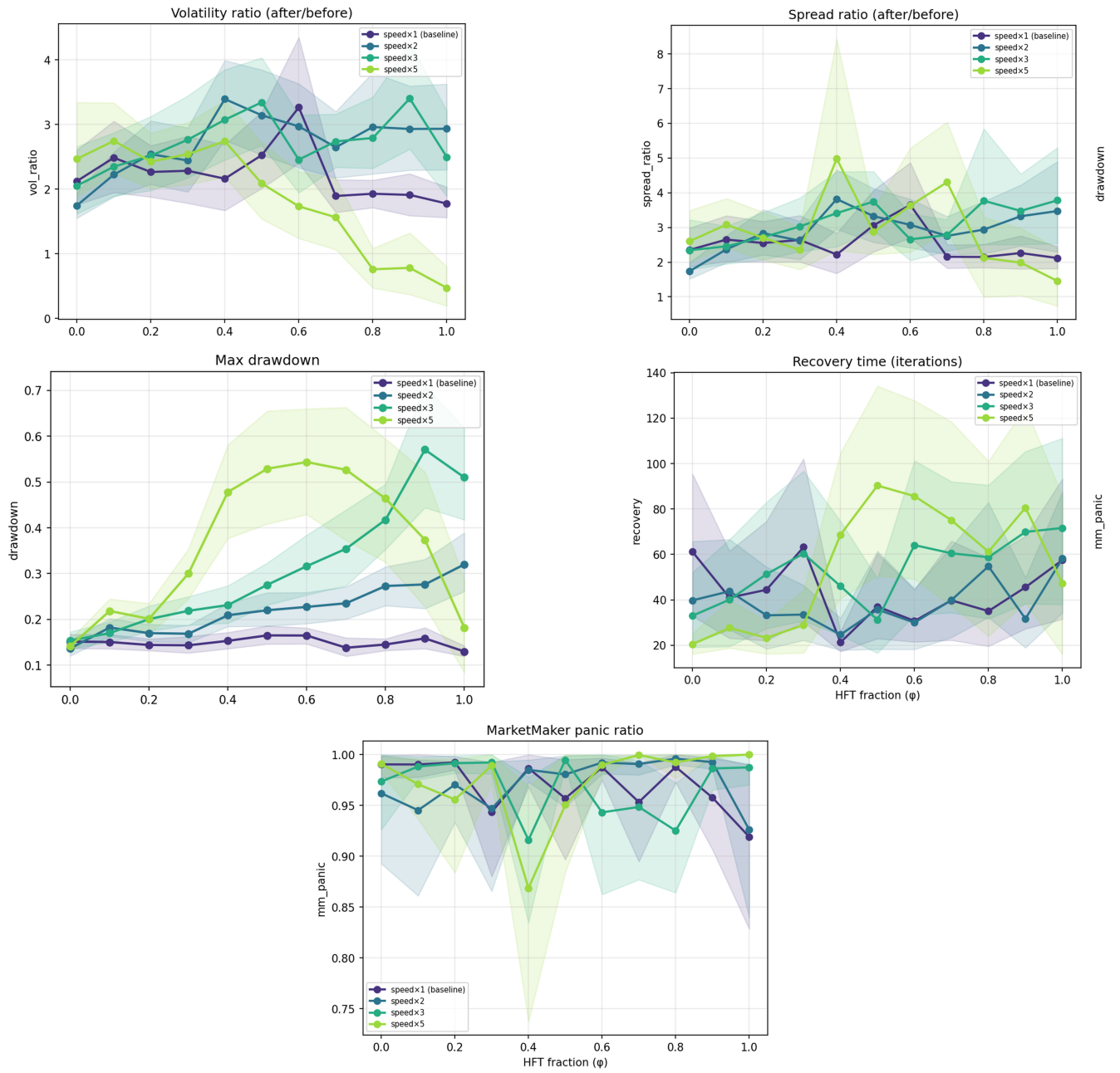


Figure 4.1 – Unified H1 speed experiment. In the speed-only branch with speed multiplier equal to 2, the tipping criterion is first met at $\phi^* = 0.2$.

The secondary H1 branches explain why the clean reported tipping result is taken from the speed-only branch. The delay-only grid does not produce a clean $1.3\times$ tipping point. Combined speed-delay effects are nonlinear and regime-dependent: delay changes the baseline instability level, so the marginal effect of speed cannot be summarised by a single threshold across all regimes. The combined speed-delay grid is shown in Figure 4.2.

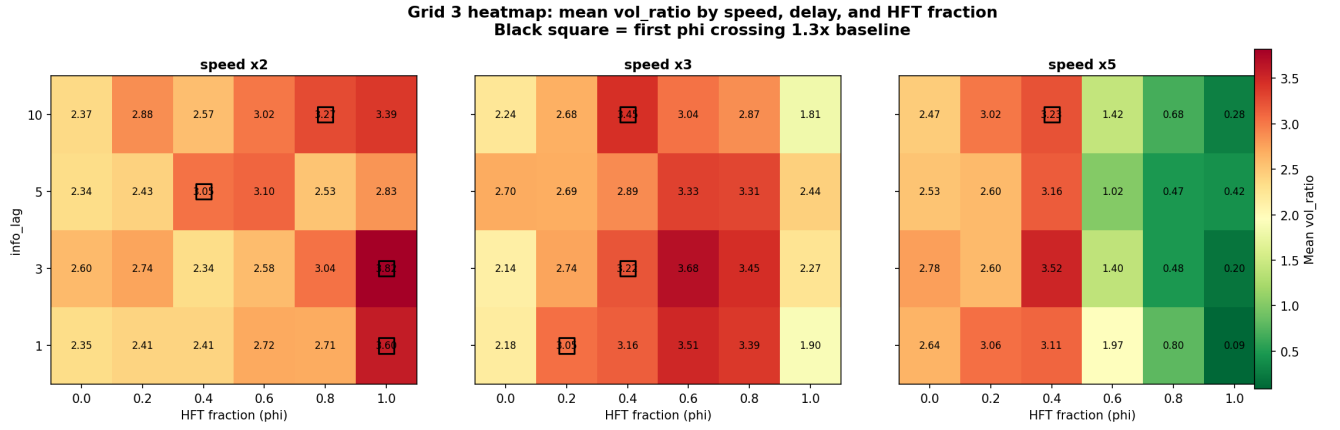


Figure 4.2 – Unified H1 combined speed-delay grid. Delay-only settings do not yield a clean tipping threshold, and the combined response depends on the regime.

The higher `speed_multiplier=5` regime is not ignored; it is retained as an extreme diagnostic case rather than used as the main H1 specification. In that regime, very frequent fast-agent trading appears to change the mechanism: the experiment no longer behaves like a clean execution-priority tipping test and instead reflects order-book absorption stress and liquidity depletion. The volatility ratio is no longer a clean monotone tipping signal, so drawdown, spread widening, recovery time, and market-maker stress become important diagnostics. For this reason, `speed_multiplier=5` is interpreted as evidence of a different extreme-stress regime, while `speed_multiplier=2` remains the main clean and interpretable H1 result.

The result supports H1 only in the final speed-multiplier-2 regime. It does not imply a monotone law: the largest mean volatility ratio occurs at an intermediate fast-agent share, and delay changes the regime.

4.2 Statistical Validation of H1

H1 is not validated only by visual inspection of the speed-grid figures and the compact H1 summary in Table 4.1. The validation asks which parts of the H1 pattern survive the operational tipping rule, threshold-sensitivity checks, non-parametric Mann–Whitney U comparisons, and bootstrap uncertainty. The answer is regime-dependent: the clearest validated result is the speed-multiplier-2 tipping pattern, not a general monotone law that higher speed always increases instability.

The tipping-rule checks support this interpretation. The threshold is defined relative to the $\phi = 0$ baseline within the same regime, so each speed or delay setting is judged against its own no-fast-chartist benchmark. Threshold sensitivity was also checked for cutoffs from $1.1\times$ to $1.5\times$ the corresponding baseline. For the clean speed-multiplier-2 regime, the crossing remains in the low-to-moderate ϕ range, which supports the interpretation that $\phi^* = 0.2$ is not merely an artefact of the arbitrary $1.3\times$ cutoff. At

the same time, the $1.3\times$ rule remains an operational criterion for a meaningful volatility increase, not a theoretical constant.

The Mann–Whitney layer narrows the support further. The strongest non-parametric evidence is concentrated mainly in the speed-multiplier-2 regime and partly in the speed-multiplier-3 regime at selected ϕ values, rather than uniformly across the whole grid. The speed-multiplier-5 regime does not survive as clean monotonic support for H1: its volatility-ratio response is weaker and non-monotonic, so it is better read as diagnostic evidence of nonlinear and extreme order-book stress.

Bootstrap confidence intervals provide the final validation layer. They show uncertainty across Monte Carlo runs and support the broad pattern that a moderate speed advantage can generate tipping behaviour under the clean H1 design. They are not used to claim that every grid point is separately decisive.

Taken together, these checks support H1 only in a regime-dependent sense. The validated claim is that a moderate execution-speed advantage among trend-following chartists can create tipping behaviour after the shock in the clean speed-multiplier-2 regime. The evidence should not be read as a general claim that more speed always destabilises the market.

4.3 H2: Calendar Time versus Event Time

H2 was initially motivated by the stronger idea that event-time measurement might reduce measured post-shock instability relative to calendar time. The final evidence is mixed, so the interpretation is narrower: event time changes how instability is measured, but it is not treated as a stabilising mechanism on its own. Section 4.3 presents the substantive results from two empirical checks: a calibrated tipping comparison across calendar time and event-time thresholds V6, V13, and V19, and a paired equal-volume comparison of calendar-time and event-time runs.

The first check uses the clean H1 benchmark as the common behavioural environment: the same speed multiplier equal to 2, information lag equal to 0, ϕ grid, shock, population, and trader rules are re-measured under calendar time and under executed-volume event-time thresholds. Calendar time reproduces the H1 clean-speed result with $\phi^* = 0.2$. V13 also gives $\phi^* = 0.2$, while V6 gives $\phi^* = 0.6$ and V19 gives $\phi^* = 0.8$. V13 is treated as the comparable event-time clock because it is closest to the baseline executed-volume scale; V6 and V19 use smaller/larger executed-volume thresholds, so they aggregate post-shock activity differently and shift the measured tipping boundary. The calibrated tipping comparison is reported in Table 4.2 and Figure 4.3.

Table 4.2 – H2 calibrated calendar-time and event-time tipping results.

Clock	Approximate executed-volume threshold	Tipping point ϕ^*
Calendar time	baseline executed volume per tick ≈ 12.915	0.2
event_time_V6	$\approx 0.5 \times$ baseline	0.6
event_time_V13	$\approx 1.0 \times$ baseline	0.2
event_time_V19	$\approx 1.5 \times$ baseline	0.8

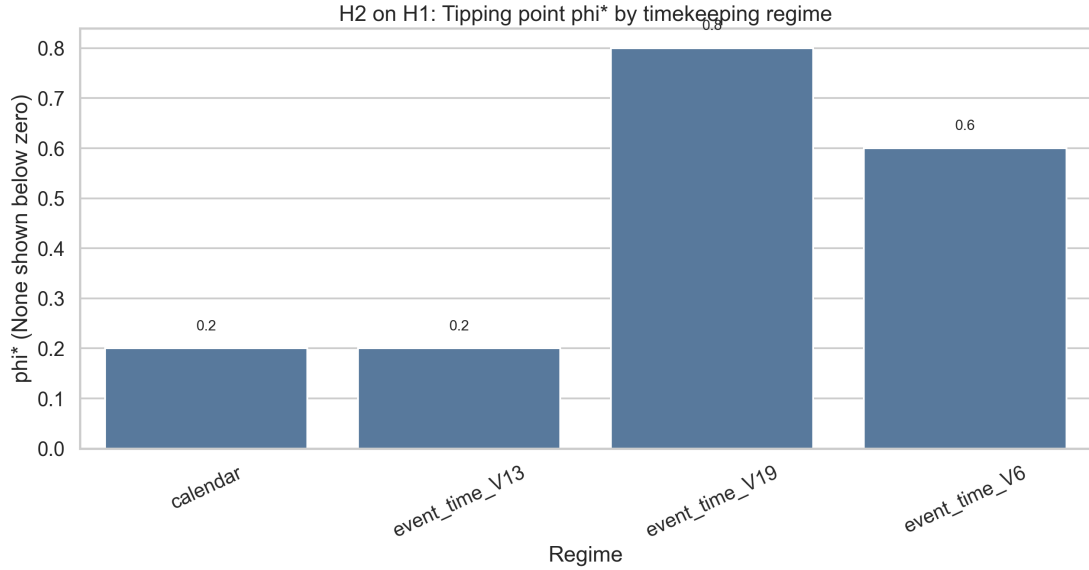


Figure 4.3 – H2 calibrated tipping comparison. The comparable event-time threshold V13 preserves $\phi^* = 0.2$, while V6 and V19 shift the crossing.

The second check is the paired equal-volume comparison. It compares calendar-time and event-time runs as paired observations with the same ϕ and seed, holding post-shock executed volume fixed across the paired observations. Substantively, this check shows that event time can lower some measured instability metrics, but only for a limited subset of outcomes. The statistical counts and correction details are reported in Section 4.4.

Together, the calibrated tipping comparison and the paired equal-volume comparison give only partial support for H2 as a measurement claim. V13 preserves $\phi^* = 0.2$, so the H1 speed result is not only a calendar-time artefact. However, V6 and V19 shift the boundary, and the paired equal-volume effect is not uniform after correction. The main implication is that clock calibration matters: event time changes measured instability but does not uniformly reduce it. Section 4.4 reports the statistical details of the paired validation.

4.4 Statistical Validation of H2

The H2 validation section focuses on how to interpret the paired equal-volume check. The paired difference is defined as the event-time metric minus the calendar-time metric, so a negative difference means lower measured instability under event time. The paired design matches ϕ and seed, while holding post-shock executed volume fixed across paired observations. This keeps the validation focused on the market-clock effect rather than on differences in the realised simulation path.

Because the validation compares many $\phi \times$ metric cells, uncorrected differences are treated as diagnostic rather than final evidence. The Wilcoxon signed-rank and sign-test results are reported with Benjamini–Hochberg and Holm corrections to distinguish broad patterns from cells that survive multiple-testing adjustment. Table 4.3 reports the corrected and opposite-direction cells that are most informative for the H2 equal-volume comparison.

Table 4.3 – H2 equal-volume paired-test cells with corrected or opposite-direction evidence.

ϕ	Metric	Mean diff. 95% bootstrap CI	Wilcoxon p	Sign p	BH p	Holm p	Direction	Survives
0.0	Calendar-style vol ratio	-0.710000 [-1.04298, -0.362293]	5.952e-05	0.001301	0.002143	0.002143	lower	Holm
0.0	Equal-volume vol ratio	-2.41750 [-4.05845, -0.835788]	0.000532	0.003300	0.009580	0.018628	lower	Holm
0.0	Spread ratio	-0.560646 [-1.01644, -0.104918]	0.002703	0.003300	0.032434	0.091895	lower	BH
0.8	Equal-volume vol ratio	-6.70018 [-14.0759, 0.187781]	0.005283	0.001301	0.047550	0.174350	lower	BH
0.2	Vol / sqrt(volume)	9.011e-05 [1.059e-05, 1.746e-04]	0.944141	0.556138	0.944141	1.000000	higher	none

Note: mean_diff is event-time metric minus calendar-time metric; for instability metrics, a negative value means lower measured instability under event time. BH/Holm significance refers to the one-sided Wilcoxon signed-rank p-values; bootstrap confidence intervals are shown as an uncertainty diagnostic and are not the sole criterion for the “Survives” column.

The strongest corrected evidence is concentrated at $\phi = 0.0$, especially in volatility-ratio metrics. This shows that event time can lower some measured instability metrics under matched-volume windows. However, corrected evidence is limited: most cells do not survive correction, and one bootstrap cell moves in the opposite direction. The paired tests therefore point to selective measurement differences rather than a general reduction in instability. The aggregate paired-test counts are summarised in Table 4.4 and Figure 4.4.

Table 4.4 – H2 paired statistical tests at equal executed volume.

Test summary	Count
Total $\phi \times$ metric cells tested	36
Uncorrected event-time-lower cells	13
Benjamini–Hochberg significant event-time-lower cells	4
Holm significant event-time-lower cells	2
Bootstrap cells where event time is higher	1

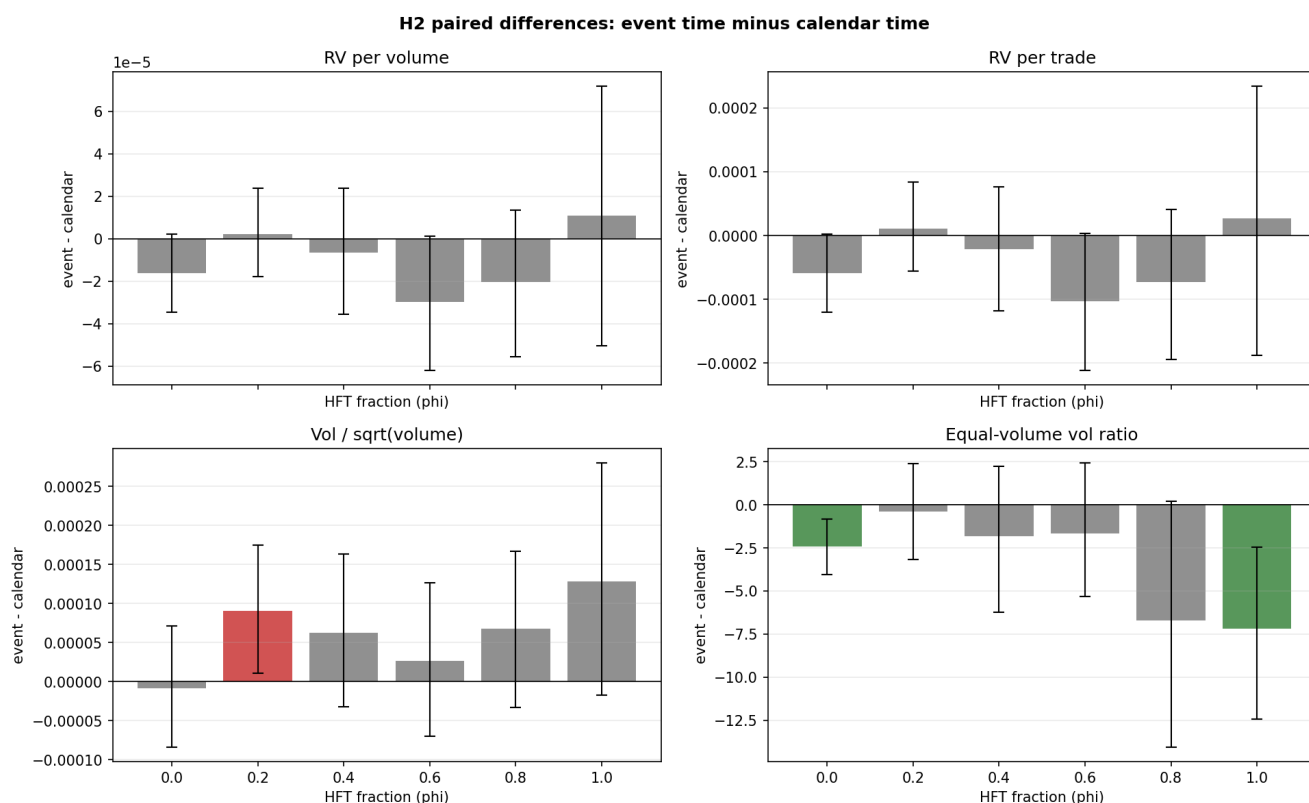


Figure 4.4 – H2 paired equal-executed-volume tests. Event time lowers some outcomes, but only a small subset of cells survives multiple-testing correction, and one bootstrap cell moves in the opposite direction.

Overall, after multiple-testing correction, the paired validation suggests that event-time aggregation can change selected instability measures, but the evidence is too narrow to support a broader stabilisation claim.

4.5 H3: Delay, Thin Liquidity, and Shock Absorption

The original H3 stated that information delay combined with illiquidity should generate stronger volatility clustering. This was the broad initial H3 test, and the volatility-clustering layer was run and evaluated rather than skipped. It is reported in the main text as a compact negative result because, across 420 volatility-clustering interaction cells, no supporting rows remain significant after Benjamini–Hochberg correction and none remain significant after Holm correction. Full diagnostic and supplementary H3 outputs are documented in Appendix B, while Table 4.5 remains the main-text summary of this negative original-layer result.

Because the broad volatility-clustering formulation has no corrected support, the H3 interpretation is narrowed ex post to the revised shock-absorption layer. This post-hoc refinement does not replace H3 with an unrelated mechanism; it narrows the same delay $\times$ thin-liquidity question from volatility persistence to shock severity and absorption. The revised claim is partially supported only through max-

imum market-price drawdown: recovery time and the stabilisation gap do not provide corrected support, and recovery time is not interpreted as stabilisation by the worst-case regime. Table 4.6 and Figure 4.5 remain the main-text evidence for the revised drawdown result.

4.6 Statistical Validation of H3

The statistical validation first records the evaluated original volatility-clustering layer in Table 4.5. The interaction test is deliberately demanding because support must survive multiple-testing correction, not only show a directionally consistent pattern. The negative corrected result rules out a robust volatility-clustering interpretation. It also motivates the narrower H3 reading introduced above; Appendix B documents the fuller diagnostic and supplementary outputs.

Table 4.5 – Original H3 volatility-clustering interaction tests.

Quantity	Value
Volatility-clustering interaction cells tested	420
Supporting rows significant after Benjamini–Hochberg	0
Supporting rows significant after Holm	0

The revised validation then applies that narrower shock-absorption reading. It tests whether delayed information combined with a thin order book weakens shock absorption after the exogenous price shock. The comparison is framed as a best-case versus worst-case contrast. The best case uses no information lag and book volume 1500; the worst case uses one-tick information lag and book volume 300.

Only maximum market-price drawdown survives correction. At $\phi = 0.2$, the worst-case mean drawdown is 0.254 compared with 0.161 in the best case, a difference of 0.093. At $\phi = 0.4$, the worst-case mean is 0.332 compared with 0.175, a difference of 0.157. Both rows remain significant after Benjamini–Hochberg and Holm correction, which makes drawdown the only robust revised-H3 outcome. Recovery time and the stabilisation gap do not survive correction, and recovery time does not provide robust adverse-direction evidence in this worst-vs-best comparison. This is not interpreted as stabilisation by the worst-case regime: in this model, deeper drawdowns can be followed by faster mechanical returns toward the recovery band, making maximum drawdown the more reliable H3 shock-absorption evidence. The revised drawdown result is reported in Table 4.6 and summarised in Figure 4.5.

Table 4.6 – Revised H3 shock-absorption result: maximum market-price drawdown.

ϕ	Best-case mean	Worst-case mean	Difference	BH-adjusted p	Holm-adjusted p
0.2	0.161	0.254	0.093	3.21612×10^{-13}	5.36021×10^{-13}
0.4	0.175	0.332	0.157	1.70899×10^{-20}	1.70899×10^{-20}

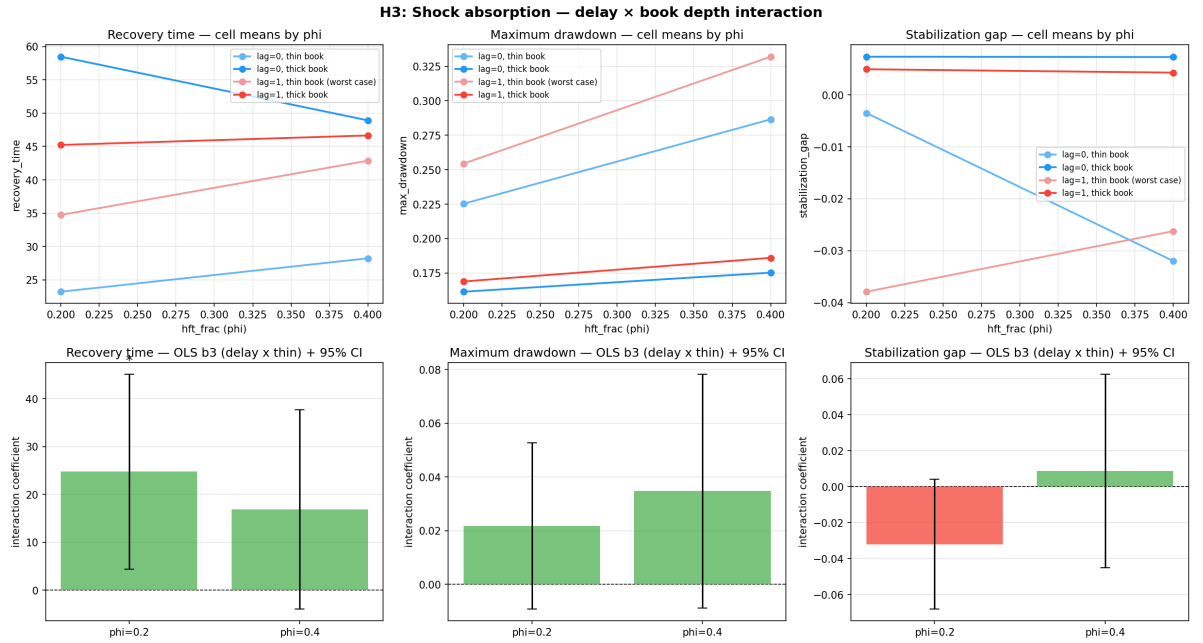


Figure 4.5 – H3 shock-absorption evidence from the delay $\times$ thin-book design. Robust support is concentrated in maximum market-price drawdown: the delayed thin-book worst case produces larger drawdowns than the no-delay deep-book benchmark at $\phi = 0.2$ and $\phi = 0.4$. Recovery time and stabilisation gap do not show the same corrected support, so H3 is interpreted as partially supported through shock severity rather than broadly supported across all shock-absorption metrics.

The statistical interpretation is therefore partial and asymmetric. H3 is not supported in its original broad volatility-clustering form, but it is partially supported in the revised shock-absorption form for maximum market-price drawdown. The evidence shows that delay under thin liquidity deepens the immediate price fall after a shock; it does not show that the same treatment reliably increases volatility clustering, extends recovery time, or creates a robust stabilisation-gap effect.

4.7 Mechanism-Level Interpretation

The three hypothesis results should be read hierarchically rather than as three equally broad stability claims. H1 is the main execution-speed hypothesis and result: it tests whether execution timing among trend-following agents can create post-shock volatility tipping. The supported interpretation is therefore mechanism-specific, not a monotone claim that more speed always destabilises the market.

H2 has a different status. It is mainly a clock-measurement and robustness check around H1,

asking whether the H1 pattern is only a calendar-time artefact. The evidence weakens that concern, but it does not introduce a separate destabilising mechanism and does not show broad event-time stabilisation. H2 therefore clarifies how the H1 pattern is observed under alternative market clocks.

H3 is the separate delay $\times$ thin-liquidity mechanism. The broad volatility-clustering formulation is not supported after correction, while the narrower shock-absorption interpretation points to maximum market-price drawdown as the robust channel. H3 therefore concerns shock severity under delayed information and thin liquidity, not broad volatility persistence.

Together, H1 and H3 describe different post-shock channels: H1 works through execution timing and volatility tipping, while H3 works through information delay under thin liquidity and drawdown severity. H2 sits between them as a measurement-clock layer around H1. This separation of mechanisms makes the mixed findings interpretable rather than contradictory and is the main contribution of the design.

4.8 Summary of Hypotheses

The final hypothesis outcomes are summarised in Table 4.7.

Table 4.7 – Summary of final hypothesis outcomes.

Hypothesis	Outcome	Interpretation
H1	Supported in the speed multiplier equal to 2 regime	Execution priority among trend-following agents is associated with a tipping point at $\phi^* = 0.2$; the maximum mean volatility ratio occurs at $\phi = 0.4$.
H2	Calendar-time artefact concern weakened	Comparable event time V13 preserves $\phi^* = 0.2$, so H1 is not only a calendar-time result; event time is regime-dependent, not consistently stabilising.
H3	Original not supported; revised partially supported	Delay $\times$ limited order-book depth does not support volatility clustering after correction, but delay $\times$ thin-order-book conditions show support for larger maximum market-price drawdown.

5 Critical Evaluation

The limitations define the domain of validity rather than weakening the research design. The model is a controlled limit-order-book laboratory rather than an empirically calibrated replica of a specific exchange. It does not make universal claims about high-frequency trading or real exchanges; it identifies

which mechanisms generate instability inside the fixed ABM environment and which broader claims fail under correction.

The representation of high-frequency trading is deliberately simplified. Fast traders are modelled as fast trend-following chartists, which captures execution-speed advantage and a crash-amplifying trend response after adverse price movement. It does not cover the full range of real HFT strategies, such as market making, latency arbitrage, order anticipation, cross-venue arbitrage, or execution algorithms. This narrows the interpretation of a “fast trader” in the project to one behavioural channel rather than the whole HFT sector.

The volatility tipping rule is also operational. The $1.3\times$ post-shock volatility threshold is a working criterion for identifying economically meaningful increases and comparing regimes on a common scale. It is not a theoretical boundary at which real markets become unstable. In this report, sensitivity to alternative thresholds from $1.1\times$ to $1.5\times$ is checked in Appendix B. Future work could extend this robustness analysis to other market designs, shock types, or empirically calibrated settings. The same applies to the 2 percent recovery band: it is an operational definition used consistently across regimes, not a structural claim about market recovery, and future robustness checks could compare 1, 2, and 3 percent bands. The noisy-delay robustness check is supplementary and is reported in Appendix B: delayed random order placement alone produces no $1.3\times$ tipping point and no significant Mann–Whitney robustness cells. This supports the mechanism-isolation interpretation: the H1 tipping pattern is not reproduced by generic delayed noisy order flow, but is tied to the fast trend-following execution-speed channel. The check remains supplementary because it does not replace a full robustness analysis over alternative noise-trader populations.

The market-clock analysis depends on event-time calibration. The calibrated V13 clock preserves the H1 boundary, while V6 and V19 shift it, so event time is best treated as a robustness and measurement framework rather than a universally stabilising clock. This limitation matters because the volume clock changes how market activity is grouped, and the interpretation of stability can depend on the selected event-time scale.

The H3 revision also limits the strength of the conclusion. The original volatility-clustering formulation was too broad, while the narrower shock-absorption framing is better supported only through maximum market-price drawdown. The project can therefore argue that delay under thin liquidity worsens one immediate shock-absorption outcome, but not that it generally lengthens recovery, changes the stabilisation gap, or creates robust volatility clustering.

The same caution applies to H2. Preserving the tipping point under the calibrated V13 event-time clock reduces the concern that H1 is only a calendar-time artefact, but the equal-volume tests support only a small corrected subset of outcomes. Event time is therefore a useful measurement lens rather than

a general market-stability mechanism in this model.

Multiple testing further constrains interpretation. Since many parameter-metric combinations are tested, correction for multiple comparisons is necessary to separate robust patterns from exploratory signals. Corrected results are treated as stronger evidence, while uncorrected patterns should remain diagnostic and useful for designing robustness checks rather than as standalone conclusions.

6 Future Work

The scope of future work follows directly from the model limitations. The current model uses one representative instrument, a stylised order book, a fixed shock design, and simplified trader rules. It does not include cross-asset contagion, venue fragmentation, strategic order cancellation, empirical calibration, or adaptive learning of HFT strategies.

A first extension would be empirical calibration to real high-frequency limit-order-book data. A future version of the model could select an intraday stress event and calibrate parameters to match observed pre- and post-shock volatility, bid-ask spreads, executed volume, maximum market-price draw-down, recovery time, book depth, and liquidity withdrawal.

A second extension would broaden the experimental design. Future work could test alternative shock types and magnitudes, different market-maker rules and inventory constraints, richer HFT strategy classes, broader robustness grids, and alternative event-time calibrations. The extreme speed-multiplier-5 result could also be studied under deeper initial book liquidity, additional market makers, stronger quote replenishment, or different inventory limits to check whether it remains an order-book absorption or liquidity-stress regime.

7 Conclusion

The paper tested three mechanisms linking trading speed, market-clock measurement, and information delay to market instability in an agent-based limit-order-book model. The outcomes are mechanism-specific: H1 is supported only in the final speed-multiplier-2 regime; H2 receives partial support as a clock-measurement robustness interpretation that weakens the calendar-time-artefact concern without showing general event-time stabilisation; and H3 is not supported in its original volatility-clustering form, but receives partial support in the revised shock-absorption form through maximum market-price drawdown.

Overall, the simulations do not show that speed, event time, or delay has one universal effect. They show that stability depends on the interaction between agent timing, market-clock measurement, and liquidity depth. The main implication is therefore mechanism-specific: careful simulation design can separate these channels and show when execution priority, clock representation, or delayed reaction affects shock propagation inside the controlled limit-order-book environment.

The final takeaway is therefore not a universal ranking of speed, event-time measurement, or delay, but a controlled account of how their effects depend on timing, measurement, and liquidity conditions.

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A Additional H2 Equal-Volume Results

The full 36-cell H2 paired-test table is stored in the public GitHub repository’s `results` branch at `results/h2/raw/h2_paired_stat_tests.csv`. The main text reports only the cells that survive multiple-testing correction or show opposite-direction bootstrap evidence, because those cells determine the substantive H2 interpretation.

The full output is included as supplementary reproducibility material rather than as a separate empirical claim. It allows the reader to trace the compact H2 summary in Section 4.4 back to the complete $\phi \times$ metric grid.

B Implementation and Reproducibility Details

The reproducibility materials are available in the public GitHub repository:

<https://github.com/ravilovaarina/market-stability-abm>

Unless stated otherwise, all file paths in Appendix B are given relative to the repository root in the `results` branch.

The project is implemented on a one-dimensional limit-order-book agent-based model with an exchange and order-book layer, heterogeneous traders, a simulator loop, event objects, and plotting and result-generation scripts. The exchange and order book are implemented through `ExchangeAgent`, `Order`, and `OrderList`. The main trader classes include `Random`, `Fundamentalist`, `Chartist`, `Universalist`, and `MarketMaker`. The simulation loop is handled by `Simulator` and `SimulatorInfo`, while shocks and other interventions use event classes such as `MarketPriceShock`, `LiquidityShock`, `InformationShock`, `MarketMakerIn`, and `MarketMakerOut`.

Implementation note: speed and information delay

HFT-like behaviour is represented through two separate mechanisms: execution priority/speed and information timing. Agents are labelled as fast or slow within the relevant experiments, and fast trend-following chartists are called before slow agents in the simulator loop.

The parameter `speed_multiplier` controls how often fast agents are allowed to trade within one simulation tick. Fast agents can receive repeated trading calls, while slow agents trade once; this represents an execution-speed advantage, not a different behavioural rule.

Information delay is represented separately through `info_lag`. Delayed agents make decisions using older price and spread observations from the recorded market-state history, which separates stale

information from execution priority. The final H1 experiments use a trend-following chartist variant because the original chartist behaviour is contrarian and can dampen price moves, whereas H1 requires a crash-amplifying fast trend-following channel after a negative shock.

H1 uses local trend-following chartist variants to represent crash-amplifying fast reaction. The unified H1 simulator calls fast agents before slow agents and gives fast agents repeated calls according to the speed multiplier. The population contains 10 fundamentalists, 10 trend-following chartists, 5 random traders, and 1 market maker. The parameter ϕ controls the chartist share assigned to the fast group. H1 includes speed-only, delay-only, and combined speed-delay branches.

H2 adds executed-volume accounting and an event-time clock, where a recorded market step ends when accumulated executed volume reaches a threshold. The final H2 design is implemented on the H1 unified environment and uses V6, V13, and V19 thresholds. The equal-volume design compares paired calendar-time and event-time runs with the same ϕ and seed, using the event-time metric minus the calendar-time metric as the paired difference.

H3 varies information delay and book depth. The final confirmatory design compares a best case with no information lag and a deep book against a worst case with one-tick information lag and a thin book. The shock-absorption scripts compute recovery time, maximum market-price drawdown, and stabilisation gap. The original volatility-clustering scripts compute autocorrelation of absolute returns, volatility persistence, and high-volatility cluster share.

The main metrics are volatility ratio, spread ratio, recovery time, maximum market-price drawdown, stabilisation gap, market-maker panic ratio, H3 clustering metrics, and H2 equal-volume metrics. The scripts use deterministic seed formulas. H1 and H2 calibrated tipping use 30 Monte Carlo runs per cell, H2 equal-volume uses 50 paired runs per ϕ , and H3 confirmatory uses 200 runs per cell. Raw CSV outputs are saved under `results/`, and tables and figures are generated from those outputs.

Supplementary robustness: noisy random-trader delay

This supplementary check applies information delay only to noisy Random agents, while trend-following chartists and fundamentalists observe current information. It tests whether stale noisy order placement alone can generate the post-shock volatility tipping pattern. The implementation and outputs are:

```
experiments/robustness/experiment_noisy_delay.py
results/h1/robustness/raw/noisy_delay_raw.csv
results/h1/robustness/tables/noisy_delay_agg.csv
results/h1/robustness/tables/noisy_delay_tipping.csv
results/h1/robustness/tables/robustness_mw_noisy_delay.csv
```

The grid varies ϕ from 0.0 to 1.0 and the noisy-agent information lag over $\{0, 1, 3, 5, 10\}$, with 30 runs per cell. The same market-price shock timing and magnitude are used as in the main experiments: $t = 200$ and $dp = -10$. The robustness check does not produce a $1.3\times$ volatility tipping point for any noisy-agent lag.

Table B.1 – Noisy random-trader delay tipping summary

info_lag	baseline_vol_ratio	threshold_1_3x	phi_star	max_vol_ratio	phi_at_max
0	1.930711	2.509925	<i>none</i>	2.197065	1.0
1	2.188088	2.844514	<i>none</i>	2.439657	0.7
3	2.453700	3.189810	<i>none</i>	2.453700	0.0
5	2.201603	2.862084	<i>none</i>	2.436742	0.7
10	2.535192	3.295750	<i>none</i>	2.552438	0.3

The additional Mann–Whitney robustness file reports 0 significant cells, 0 above-threshold cells, and 0 tipping-point cells. This check is supplementary and should not be treated as a separate hypothesis. It supports the mechanism-isolation interpretation: delaying noisy Random-agent order placement alone does not reproduce H1 tipping behaviour. The main speed-driven result therefore appears tied to the fast trend-following mechanism rather than to generic delayed noisy order flow.

Supplementary robustness: H1 threshold, shock magnitude, and non-parametric checks

This appendix reports additional robustness checks for H1. First, the tipping-point threshold is varied from $1.1\times$ to $1.5\times$ of the baseline volatility ratio. Second, the exogenous shock magnitude is varied while holding `speed_multiplier=2` and `info_lag=0` fixed. Third, one-sided Mann–Whitney U tests compare volatility ratios across speed and delay regimes at fixed ϕ . These checks are supplementary: they do not replace the main H1 result, but they test whether the speed-driven tipping interpretation is sensitive to operational choices. The implementation and outputs are:

```
experiments/robustness/experiment_threshold_validation.py
experiments/robustness/experiment_shock_magnitude.py
experiments/robustness/experiment_speed_delay_mannwhitney.py
results/h1/robustness/tables/threshold_validation_summary.csv
results/h1/robustness/tables/shock_magnitude_tipping.csv
results/h1/robustness/tables/speed_effect_mannwhitney.csv
results/h1/robustness/tables/delay_effect_mannwhitney.csv
```

Table B.2 – H1 threshold-sensitivity check for alternative tipping definitions.

Regime	Baseline vol. ratio	ϕ^* at $1.1\times$	ϕ^* at $1.2\times$	ϕ^* at $1.3\times$	ϕ^* at $1.4\times$	ϕ^* at $1.5\times$	Note
speed_x2	1.745992	0.1	0.1	0.2	0.2	0.4	Key clean speed regime
combined_speed_x3_lag1	2.178019	0.2	0.2	0.2	0.2	0.6	Strong combined regime with stable tipping
combined_speed_x3_lag5	2.699817	0.6	0.6	none	none	none	Combined regime where $1.3\times$ tipping disappears

The clean speed-multiplier-2 regime retains a finite tipping point across all tested threshold definitions. The estimated boundary moves from $\phi = 0.1$ under lenient thresholds to $\phi = 0.4$ under the strictest threshold, so the precise boundary is threshold-sensitive, but the existence of a speed-driven transition is not created solely by the $1.3\times$ rule. The combined regimes are less stable, which is consistent with the main interpretation that speed and delay interact nonlinearly.

Table B.3 – H1 shock-magnitude robustness under speed-multiplier-2.

Shock dp	Baseline vol. ratio	Threshold $1.3\times$	ϕ^*	Max vol. ratio	ϕ at max
-0.5	1.726233	2.244103	0.4	3.185704	1.0
-1.0	1.726465	2.244404	0.4	3.172642	0.9
-2.0	1.760541	2.288704	0.3	3.136494	1.0
-5.0	1.990777	2.588010	0.7	3.260022	0.8
-10.0	2.312319	3.006015	0.5	3.721872	0.9

The shock-magnitude robustness grid is generated by a separate robustness script with an independent deterministic seed ensemble. Therefore, the $dp = -10$ baseline in Table B.3 is not expected to be numerically identical to the main H1 baseline in Table 4.1, even though the parameter setting is comparable. The table is used to assess robustness across shock magnitudes, not to reproduce the exact Monte Carlo sample of the main H1 run.

A tipping point is present for every tested shock magnitude, with ϕ^* ranging from 0.3 to 0.7. This means the speed-driven tipping pattern is not tied to a single shock calibration. At the same time, the location of the threshold varies with shock size, so the result should still be interpreted as regime-dependent rather than universal.

The Mann–Whitney speed checks show stronger evidence for the speed mechanism than the delay-only mechanism. For speed comparisons against the speed-multiplier-1 baseline, significant increases appear at selected medium and high ϕ values, especially for speed multipliers 2 and 3; speed multiplier 5 is not consistently significant and should not be interpreted as monotonic support. Delay-only comparisons are weaker: significant rows are concentrated at $\phi = 0.8$ for lags 3, 5, and 10, while most delay rows are not significant. These tests use uncorrected $p < 0.05$, so they should be treated as supporting diagnostics rather than final corrected evidence.

Table B.4 – Summary of supplementary non-parametric H1 robustness checks.

Check	Strongest supported cells	Weak/non-supported part	Interpretation
Speed effect	speed_multiplier 2 and 3 at selected ϕ values, especially ϕ 0.4, 0.8, 1.0	speed_multiplier 5 is not consistently significant	Supports regime-dependent speed effect, not monotonic speed effect
Delay effect	Significant rows mainly at ϕ 0.8 for lags 3, 5, 10	Most delay rows are not significant	Delay-only effect is weaker and less stable
Noisy random-agent delay	0 significant cells, 0 above-threshold cells, 0 tipping cells	No noisy lag creates tipping	Random-agent delay alone does not reproduce H1

Supplementary H3 analysis: clustering, depth, and shock absorption

H3 was evaluated in several layers. The original layer tested volatility clustering using autocorrelation of absolute returns, volatility persistence, and high-volatility cluster share. The liquidity-depth grid then varied book volume to test whether thin liquidity strengthened delay effects. Because the broad clustering formulation did not survive correction, the final confirmatory design reframed H3 as shock absorption and compared best-case versus worst-case regimes. The implementation and outputs are:

```
experiments/h3/experiment_h3_volatility_clustering.py
experiments/h3/experiment_h3_liquidity_depth.py
experiments/h3/experiment_h3_confirmatory_2x2.py
experiments/h3/experiment_h3_shock_absorption.py
experiments/h3/experiment_h3_final_summary.py
results/h3/clustering/
results/h3/depth/
results/h3/confirmatory/
results/h3/shock_absorption/
results/h3/summary/
```

Table B.5 – Supplementary H3 experimental layers.

H3 layer	Design grid	Runs/data	Purpose	Final use
Original clustering grid	hft_frac 0.0 to 1.0, info_lag 0/1/3/5/10, softlimit 20/50/100	2700 raw rows	Tests broad volatility-clustering interaction	No corrected support
Liquidity-depth grid	hft_frac 0.0/0.2/0.4/0.6, info_lag 0/1/3/5, book_volume 300/600/1000/1500	1920 raw rows	Tests delay under different book depths	Mixed exploratory evidence
Confirmatory 2x2	info_lag 0/1, book_volume 300/1500, ϕ 0.2/0.4	1600 raw rows	Focused shock-absorption design	Used for final H3 interpretation
Worst-vs-best shock absorption	Best: lag 0 and book 1500; worst: lag 1 and book 300	Analysis from confirmatory raw	Direct contrast of shock absorption	Robust only for max drawdown

Table B.6 – H3 confirmatory 2x2 shock-absorption interaction summary.

ϕ	Metric	Baseline	Delay only	Thin book only	Combined	Interaction
0.2	Recovery time	58.460	45.235	23.235	34.750	24.740
0.2	Max drawdown	0.161	0.169	0.225	0.254	0.022
0.2	Stabilisation gap	0.007	0.005	-0.003	-0.038	-0.032
0.4	Recovery time	48.915	46.655	28.240	42.870	16.890
0.4	Max drawdown	0.175	0.186	0.287	0.332	0.035
0.4	Stabilisation gap	0.007	0.004	-0.032	-0.026	0.009

Table B.7 – H3 worst-vs-best shock-absorption comparison.

ϕ	Metric	Best-case mean	Worst-case mean	Difference	p -value	Significant
0.2	Recovery time	58.460	34.750	-23.710	1.000	no
0.2	Max drawdown	0.161	0.254	0.093	1.072e-13	yes
0.2	Stabilisation gap	0.007	-0.038	-0.045	0.989	no
0.4	Recovery time	48.915	42.870	-6.045	1.000	no
0.4	Max drawdown	0.175	0.332	0.157	2.848e-21	yes
0.4	Stabilisation gap	0.007	-0.026	-0.034	0.776	no

Note: In Table B.6, the “Combined” cell is the delay + thin-book treatment. In Table B.7, the “worst case” is the same delay + thin-book combination, while the “best case” is no delay + deep book.

The supplementary H3 analysis confirms that the original broad volatility-clustering claim is too strong. The final corrected summary contains 420 clustering-interaction tests, and no row survives Benjamini–Hochberg or Holm correction at the 5 percent level. The shock-absorption reframing is narrower. In the worst-vs-best comparison, only maximum market-price drawdown is robustly worse in the delayed thin-book regime, at both $\phi = 0.2$ and $\phi = 0.4$. Recovery time and stabilisation gap do not survive correction, and recovery time does not provide robust adverse-direction evidence in this worst-vs-best comparison. Therefore, H3 should be interpreted as partial evidence that delay under thin liquidity deepens the immediate price fall after a shock, not as evidence of broad volatility clustering or generally slower recovery.

Experiment inventory

This appendix inventory summarises the main experiment families implemented in the project. It is included to document the computational scope of the work and to separate the core hypothesis tests from supplementary robustness checks.

Table B.8 – Experiment inventory.

Experiment family	Main scripts / outputs	Design role	Included in report as
H1 speed-only grid	experiments/h1/experiment_unified.py; results/h1/raw/unified_speed_raw.csv	Tests whether execution-speed heterogeneity creates a post-shock volatility tipping point.	Main H1 result and validation.
H1 delay-only grid	results/h1/raw/unified_delay_raw.csv	Tests whether information delay alone creates a clean tipping point.	H1 diagnostic / secondary result.
H1 combined speed-delay grid	results/h1/raw/unified_combined_raw.csv	Tests nonlinear interaction between speed and delay.	H1 secondary result and robustness discussion.
H1 robustness checks	results/h1/robustness/tables/threshold_validation_summary.csv; results/h1/robustness/tables/shock_magnitude_tipping.csv; results/h1/robustness/tables/speed_effect_mannwhitney.csv; results/h1/robustness/tables/delay_effect_mannwhitney.csv	Tests threshold sensitivity, shock-size sensitivity, and non-parametric support.	Appendix H1 robustness.
Noisy random-trader delay	experiments/robustness/experiment_noisy_delay.py; results/h1/robustness/tables/noisy_delay_tipping.csv; results/h1/robustness/tables/robustness_mw_noisy_delay.csv	Tests whether delayed random order flow alone reproduces H1 tipping.	Supplementary appendix robustness.
H2 standalone volume-clock prototype	h2-event-time-volume-clock	Develops the event-time mechanism and identifies why final H2 must be matched to H1.	Methodological background, not final evidence.
H2-on-H1 calibrated tipping	results/h2/raw/h2_unified_calibrated_tipping.csv	Tests whether H1 tipping survives calendar versus event-time measurement.	Main H2 result.
H2 equal-volume paired tests	results/h2/raw/h2_paired_stat_tests.csv; results/h2/raw/h2_postshock_volume_paired_diffs.csv	Compares calendar and event time under matched post-shock executed volume.	H2 statistical validation and appendix note.
H3 original volatility-clustering grid	results/h3/clustering/raw/h3_clustering_raw.csv; results/h3/summary/tables/h3_final_summary_clustering_interactions.csv	Tests original delay $\times$ liquidity volatility-clustering claim.	Rejected original H3 formulation.
H3 liquidity-depth and confirmatory grids	results/h3/depth/raw/h3_depth_raw.csv; results/h3/confirmatory/raw/h3_confirmatory_raw.csv	Tests delay-depth interaction and focused shock-absorption design.	H3 supplementary appendix.
H3 shock absorption / final summary	results/h3/shock_absorption/tables/h3_shock_absorption_mw.csv; results/h3/summary/tables/h3_final_summary_shock_absorption.csv	Tests best-case versus worst-case shock absorption.	Final H3 evidence.

The inventory shows that the final claims are based on a sequence of core simulations, diagnostic checks, and supplementary robustness tests rather than on a single isolated run.

Metric definitions and statistical tests

This appendix subsection summarises the metrics and statistical tests used across the experiments. The aim is to make clear how the stability outcomes in the main text are operationalised and how simulation uncertainty is assessed.

Table B.9 – Metric definitions and statistical tests used in the experiments.

Metric / test	Definition or role	Used mainly in	Interpretation
Volatility ratio	Post-shock price volatility divided by the pre-shock price-volatility benchmark.	H1 and H2 tipping analysis.	Higher values mean stronger post-shock instability relative to the pre-shock regime.
Spread ratio	Post-shock relative spread divided by pre-shock relative spread.	Liquidity and post-shock market-quality checks.	Values above one indicate deterioration in quoted liquidity.
Maximum market-price drawdown	Largest relative fall from the pre-shock market price to the lowest post-shock market price.	H3 shock absorption.	Higher values mean deeper immediate price damage after the shock.
Recovery time	Number of ticks until price enters the 2 percent band around the pre-shock level; if no recovery occurs, the remaining horizon is used.	H1 and H3 shock absorption.	Higher values mean slower recovery.
Stabilisation gap	Distance between the final post-shock stabilisation price and the pre-shock price.	H3 shock absorption.	Higher positive values mean the market settles further below the pre-shock level.
Market-maker panic ratio	Share of post-shock ticks where the market maker cannot provide normal two-sided liquidity.	Market-maker stress checks.	Higher values indicate more frequent liquidity-provision failure.
Equal-volume metrics	Realised variation per executed volume, realised variation per trade, volatility scaled by square root of volume, equal-volume volatility ratio, and calendar-style volatility ratio.	H2 equal-volume comparison.	Separate measured instability from differences in trading intensity.
Volatility-clustering metrics	Autocorrelation of absolute returns at lags 1 and 5, volatility persistence, and high-volatility cluster share.	Original H3 formulation.	Measure whether large moves or high volatility persist after the shock.
Mann–Whitney U test	Non-parametric comparison between treatment and baseline distributions.	H1 and H3 robustness.	Useful when Monte Carlo outputs are non-normal or affected by outliers.
Wilcoxon signed-rank test	Non-parametric paired test for matched differences.	H2 equal-volume paired tests.	Tests whether event-time and calendar-time paired differences are systematically negative or positive.
Sign test	Paired direction test based only on the sign of differences.	H2 equal-volume paired tests.	Checks whether negative differences are more frequent than positive differences.
Bootstrap confidence intervals	Resampling-based uncertainty intervals for mean differences and plotted metrics.	H1/H2/H3 uncertainty reporting.	Shows Monte Carlo uncertainty without relying on normality.
OLS with HC3 robust standard errors	Regression-based interaction test with heteroskedasticity-robust errors.	H3 delay $\times$ thin-book interaction checks.	Tests whether the combined treatment differs from additive main effects.
Benjamini–Hochberg and Holm corrections	Multiple-testing corrections across many parameter-metric cells.	H2 and H3 validation.	Corrected results are stronger evidence; uncorrected results are treated as diagnostic.

The table is intended as a reader guide. The main text reports the substantive results, while this appendix clarifies how the reported metrics and tests should be interpreted across the simulation experiments.

Reproducibility and output map

The full implementation, experiment scripts, generated result tables, and figure-generation code are available in the project repository:

<https://github.com/ravilovaarina/market-stability-abm>.

The final H1, H2, and H3 outputs referenced below are collected in the public repository’s `results` branch.

Replication protocol

- 1 Clone the GitHub repository linked above.
- 2 Install the project dependencies listed in the repository documentation.

- 3 Run or inspect the H1, H2, and H3 experiment scripts to verify the simulation design.
- 4 Regenerate the raw CSV outputs under `results/`.
- 5 Regenerate the derived tables and figures from the saved outputs.
- 6 Compare the generated figures and tables with the corresponding report outputs.
- 7 Use the deterministic seed formulas in the scripts to reproduce the Monte Carlo grids.

The project is organised so that simulations, raw outputs, derived tables, and figures can be traced back to experiment scripts. This section records the main output locations used to support the reported results.

Table B.10 – Reproducibility and output map.

Result block	Raw outputs / tables	Figures / report output	Reproducibility note
H1 main results	<code>results/h1/raw/unified_speed_raw.csv</code> ; <code>results/h1/raw/unified_delay_raw.csv</code> ; <code>results/h1/raw/unified_combined_raw.csv</code> <code>results/h1/robustness/tables/</code>	H1 speed/delay/combined figures and summary tables.	Generated by unified H1 experiment with deterministic seed formulas and 30 runs per cell.
H1 robustness		Appendix threshold, shock, noisy-delay, and Mann–Whitney tables. Calendar versus event-time tipping comparison.	Robustness scripts reuse saved H1 outputs or run fixed robustness grids. Final H2 is matched to the clean H1 environment.
H2 calibrated tipping	<code>results/h2/raw/h2_unified_calibrated_tipping.csv</code>		
H2 equal-volume paired tests	<code>results/h2/raw/h2_postshock_volume_paired_diffs.csv</code> ; <code>results/h2/raw/h2_paired_stat_tests.csv</code> <code>results/h3/clustering/</code> ; <code>results/h3/depth/</code>	H2 compact equal-volume table and paired-test figure.	Calendar/event-time runs use matched ϕ and seed pairs.
H3 clustering/depth		Original H3 and exploratory depth evidence.	Used to test and then reject the broad volatility-clustering formulation.
H3 confirmatory/shock absorption	<code>results/h3/confirmatory/</code> ; <code>results/h3/shock_absorption/</code> ; <code>results/h3/summary/</code>	Final H3 drawdown evidence and appendix tables.	Confirmatory 2x2 and worst-vs-best tests support only maximum drawdown.

The report therefore separates simulation generation, statistical analysis, and presentation. The appendix does not introduce new empirical claims; it documents how the claims in the main text are connected to saved outputs and reproducible scripts.

C Use of LLM Assistance

Large language model assistance was used during the project as a consultative tool for clarification, checking, and editing support. The main tools used were Claude (Anthropic, claude.ai) and ChatGPT (OpenAI), accessed during January–May 2026. The models were not used to generate simulation results. The experiment design, code execution, numerical outputs, interpretation of results, and final decisions about the report remained the author’s responsibility.

The assistance was used in four main ways. First, LLMs were used to clarify terminology from market microstructure and statistics, such as the difference between calendar time and event time, the meaning of flow toxicity, the interpretation of bootstrap confidence intervals, and the difference between false discovery rate and familywise error rate correction. Second, they were used to check whether implementation choices matched the intended economic logic, for example when separating execution speed from information delay in the simulator. Third, they were used for sanity-checking the interpretation of counterintuitive simulation results. Fourth, they were used for language polishing and formatting advice, while the substantive claims and numerical results were checked against the project outputs.

Table C.1 gives examples of prompts and model responses used during the work.

Table C.1 – Examples of LLM assistance used in the project.

Prompt	LLM tool	Summary of response	How the response was used
“Explain the difference between calendar time and event time in high-frequency market data. In my simulation, calendar time records every tick, while event time records observations after a fixed amount of executed volume. How should I describe this difference in a research report?”	ChatGPT (OpenAI)	The model explained that calendar time uses equal clock or simulation intervals, while event time groups observations by market activity. It also noted that event time can change measured volatility because active and inactive periods are weighted differently.	Used to clarify the H2 motivation and to describe the event-time comparison as a measurement-clock question rather than as an automatic stabilisation mechanism.
“In an ABM with a limit order book, I separate fast and slow agents. Fast agents are called before slow agents, and <code>speed_multiplier</code> gives fast agents repeated trading calls within one tick. Is this better described as execution speed or information speed?”	Claude (Anthropic)	The model distinguished execution priority from information timing. It suggested describing repeated calls and earlier activation as execution-speed advantage, while delayed price/spread observations should be treated as information delay.	Used to make the methodology clearer and to separate the H1 execution-speed mechanism from the H3 information-delay mechanism.
“This paragraph says that H3 is supported because maximum drawdown is larger in the delayed thin-book regime, while recovery time and stabilisation gap do not survive correction. Does the wording overstate the evidence? Please suggest a more cautious phrasing without changing the numerical result.”	ChatGPT (OpenAI)	The model suggested avoiding a broad statement that H3 is fully supported. It recommended phrasing the result as partial support concentrated in maximum market-price drawdown, while explicitly saying that recovery time, stabilisation gap, and the original volatility-clustering formulation are not robustly supported.	Used to make the H3 wording more cautious and avoid overstating the evidence. The numerical values, corrected <i>p</i> -values, and final interpretation were checked against the project’s simulation outputs and remained the author’s responsibility.

All LLM-generated suggestions were checked before being included in the report. The models were used as an auxiliary discussion and editing tool, not as a substitute for running experiments,

analysing outputs, or making final research decisions.